**ISMAEL VILLEGAS MOLINA**

3775 Blossom Way, El Centro, CA 92243 | 760-457-7205 | ismaelvillegas3@gmail.com | ismael.villegas@usc.edu

## EDUCATION

University of Southern California | Viterbi School of Engineering — Los Angeles, CA
**M.S. in Computer Science** — **May 2021**

Columbia University in the City of New York | The Fu Foundation of Engineering and Applied Science — New York, NY
**B.S. in Computer Science** — **May 2019**
Emphasis in Intelligent Systems

## AWARDS

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|---|---|
| Gates Millennium Scholar | 2015 – 2025 |
| GEM Fellow | 2019 – 2021 |
| Hispanic Scholarship Fund Scholar | 2015 – 2019 |
| Norman A. Schefer Scholar, Columbia University | 2015 – 2019 |
| Dean's List | 2019 |

## RELATED EXPERIENCE

University of Southern California – Institute of Creative Technologies — Los Angeles, CA
**Graduate Research Assistant (Under Prof. Gale Lucas)** — **Sept. 2019 – Present**
Proposed and carried out an experiment where we toggle the language formality on a Virtual Human and gather data on the human perception of these machines. Paper under review at CHI conference. Future work from proposal includes implementing Spanish-English codeswitching and African American Vernacular English to diversify the language used by Virtual Humans. We would also like to implement a Text-to-Speech capability to see the differences in perception of formality levels rather than only written language.

Amazon — Seattle, WA
**Applied Science Research Intern (AWS Lex Science Team | Customer Service ML Team)** — **Summer 2019; Summer 2020**
Part of the Amazon Web Services', Lex Science Team. Worked on data augmentation methods to create a better chatbot developing experience for customers. Developed a self-training approach where we start with small, annotated data seeds that we use to train a machine learning model. We use that model to create predictions on unannotated data, then use the original annotations and these newly predicted annotations to create another machine learning model that shows high performance with small amount of gold standard data. | Part of Amazon's Customer Service ML Team. Created the answer extraction model for a question-answering pipeline. Given a question and a document that contains an answer to that question, extract the correct answer. Used various ML architectures such as BERT and XLNet. Used data augmentation techniques to improve on initial dataset.

Columbia University Speech Lab — New York, NY
**Research Assistant (Under former Dept. Chair of Computer Science Prof. Julia Hirschberg)** — **Sept. 2017 – May 2018; Jan. 2019 – May 2019**
Studying the neuro-phenomenon of English-Spanish code-switching. Using an EEG to check neural activity of English-Spanish bilinguals when code-switching occurs in conversation. Analyze if a confederate could elicit code-switching from a subject. Created game for subjects to play with confederate in order to elicit speech for data collection. Analyze comprehension and production of spontaneous English-Spanish code-switching. Used LIWC software to analyze code-switched speech and see if there is a correlation between switched speech and personal topics/informal speech. Certified for training in CITI Conflict of Interest for PHS Researchers, Human Subjects Protection includes FDA, Responsible Conduct of Research (RCR), and How to do Human Subjects Protocols. | Take scraped online, bilingual data and determine if any two given files belong together as parallel data. Perform word alignments for Somali-English and Swahili-English parallel documents.

Columbia University Bauer Lab — New York, NY
**Research Assistant (Under Dr. Daniel Bauer)** — **Jan. 2019 – May 2019**
Data collection done for the course "Introduction to Computing for Engineers and Applied Scientists" to start tailoring a new form of lecturing. Trying to find a way to encourage aspiring computer scientists to participate with their peers without being discouraged from other, much more experienced computer science students. Along with this, balance the course to not hold back those with past experience.

Columbia University Natural Language Processing Lab — New York, NY
**Research Assistant (Under Prof. Kathleen McKeown)** — **Sept. 2018 – Dec. 2018**
In collaboration with the Electrical Engineering Department. Scrape data online to find energy tips. Build data set to show why the certain reasoning was used in the tip. Automatically produce readable energy tips found online in various forms (simple statement, question, etc.) based on the context of the tips found. Verify newly formed tips for energy efficiency accuracy and give to participants. Monitor if tips are being used and automatically change framing of tip to promote use. Programming in Python.

Columbia University Media and Idea Lab — New York, NY
**Research Assistant (Under Prof. Frances Negrón-Muntaner)** — **Sept. 2018 – Present**
Working under the Comparative Literature Department. Scrape web data regarding Latinxs in Comedy. Build data set of directors, producers, and writers who are of Latinx descent from the past half century. This is to write a report to determine what part Latinxs have played in the making of the highest rated comedies behind the scenes. Programming in Python.

Human-Assistive Robotic Technologies Lab at UC Berkeley — Berkeley, CA
**Research Assistant (Under Prof. Ruzena Bajcsy)** — **Summer 2018**
Part of Smart Vehicles project that worked with semiautonomous systems to improve a human's comfort. Studying how the psychoacoustics affect the focus of a driver. Compute and analyze the annoyance levels of sounds inside a self-driving car and adapt to the situation to lower annoyance levels for driver. Programming done in C++ and Python. Enhanced previous framework on Grand Theft Auto V simulator. Gathered real-life data with a Lincoln MKZ equipped with the following instruments: eight webcams, LiDAR sensor, GPS, Inertial Measurement Unit, CAN bus, GoPro Hero 5, and two Olympus microphones. Performed in partnership with Computing Research Association (CRA) through their DREU program. Work found here: https://ismaelvillegasdreu2018.wordpress.com/

Goldman, Sachs & Co. — New York, NY
**Summer Analyst (Securities Division – FICC Technology)** — **Summer 2017**
Part of Fixed Income, Currencies, and Commodities (FICC) technology team. Worked primarily in Java. Created JUnit tests for existing programs. Worked on post-trade processing. Worked with various venues (Tradeweb, Bloomberg, Brokertec) to process trades made by European Repo Desk. Used JMS with EMS to connect and receive messages from trades all from scratch. This was done to increase transparency of markets, lower cost of market data, and have more explicit costs of trading and investing securities.

Dynamic Consulting Engineers — Imperial, CA
**Civil Engineering Intern** — **Summer 2016**
Worked with AutoCAD to design ALTA maps. Recording elevations and setting targets for pictures to record GPS coordinates with Trimble R8. Improved website using HTML. Provided assistance with the bidding process of new projects throughout the Imperial County.

Imperial County Veterans' Memorial — Imperial, CA
**Lead Volunteer** — **2013 – 2017**
Team member for design and construction of an 8,400ft$^2$ Veterans' Memorial with colored sidewalks, an eagle-shaped 55' by 8' concrete wall with six 18" military seals, three flagpoles, five light poles and benches. Helped with permitting and presentations with local government agencies. Assisted on site with digging, compaction, and cleaning.

## PUBLICATIONS AND PRESENTATIONS

*"Yo That's a Sick Deal!": Perception of Colloquial Communication with Virtual Humans*
**Ismael Villegas**, Johnathan Mell, Kushal Chawla, Gale Lucas
Paper under review at ACM CHI Conference on Human Factors in Computing Systems — **2020**

*On the Development of an Acoustic-Driven Method to Improve Driver's Comfort Based on Deep Reinforcement Learning*
Erickson R. Nascimento, Ruzena Bajcsy, Michal Gregor, Isabella Huang, **Ismael Villegas**, Gregorij Kurillo
Journal publication in IEEE Transactions on Intelligent Transportation Systems — **2020**

*Acoustic-driven Interior Vehicle Adaptation based on Deep Reinforcement Learning to Improve Driver's Comfort*
Erickson R. Nascimento, Ruzena Bajcsy, Gregorij Kurillo, **Ismael Villegas**

Paper and poster at International Conference on Intelligent Robots (IROS) workshop, Madrid, Spain **2018**

*"Quiet Down!": A Study of Psychoacoustic Annoyance Within Vehicles*
**Ismael Villegas**, Erickson R. Nascimento, Ruzena Bajcsy
Poster at MIT Undergraduate Research Technology Conference, Cambridge, MA, USA **2018**

## PROJECTS

Imperial Valley Upward Bound Computer Science Program — Imperial Valley, CA
**Lead, Founder** — **2018 – Present**
Starting computer science program for high school students in the Imperial Valley in California. Little to no computer science education available in the area and program promotes programming and thinking like a computer scientist. Goal of program is to provide CS for students, but ultimately have established CS courses in all (7) public high schools in the area.

Web Technologies — Los Angeles, CA; New York, NY
**Lead** — **2019 – 2020**
Implemented a full-stack news app using 5+ APIs including Python/Node backends, HTML/CSS, Bootstrap React, Android based clients with GCP deployment. This was created in both a website and mobile application. | Created a personal playlist creator based on user's top listened songs from past months using the Spotify API.

Natural Language Processing (COMS 4705) — New York, NY
**Lead** — **2018**
Created n-gram language models with smoothed probabilities and used the model for text classification. Used CKY algorithm for CFG and PCFG parsing and produce a parse tree. Used WordNet and Word2Vec word embeddings for lexical substitution, trained neural network to predict transitions in arc-standard dependency parser. All written in Python.

Artificial Intelligence (COMS 4701) — New York, NY
**Lead** — **2017**
Wrote program to solve sliding n-puzzles using Breadth-First-Search, Depth-First-Search, Greedy-Best-Search, and the A* algorithm while using Manhattan distance or misplaced node as heuristic. Created an AI player for the game Reversi using Minimax or alpha-beta pruning and depth limits. Created a spam filtering system on text message dataset and a Naïve Bayes classifier. Built a neural network for image classification. All written in Python.

Advanced Programming (COMS 3157) — New York, NY
**Lead** — **2017**
Wrote a database lookup network server using sockets API, wrote an HTTP client (a web page downloader), wrote a web server from scratch (serving static pages, serving dynamic pages). All written in C.

Voice-controlled Robot — Cambridge, MA
**Contestant** — **2016**
Created a robot that is voice controlled over Wi-Fi using the Microsoft's Bing Speech to Text API. Built using the Intel Edison and Arduino Software. Final idea would be to create robot that elder people could use to transport pills around a house when people cannot move as much. Part of the HackMIT competition.

## SKILLS

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|---|---|---|---|
| Java | C/C++ | Python | ROS |
| HTML/CSS | LaTeX | Matlab | Javascript/JQuery |

## LANGUAGES

English – Native Language
Spanish – Native Language
French – Speak, read, and write with basic proficiency

## ORGANIZATIONS

**Jopes Foundation** *(10/2010) – Founder and Co-President*. A non-profit organization that aims to help underprivileged orphans in Mexico with an emphasis on Mexicali, BC and Tijuana, BC.
**Global Brigades, Water/Public Health Chapters, Columbia University** – *Vice President, Internal and External Education Chair, and Elected Officer*. Global group that visits developing countries that empowers communities to meet with health and economic goals through volunteers.
**Society of Hispanic Professional Engineers (SHPE), Columbia University** – *Vice President, Treasurer*. Prepare for renowned national conference and participate in skill building activities for interviews and networking events. Allocate and keep track of funds of the group. Working with school officials to fund travel to national conference and weekly meetings.
**IBM Mentorship Program** – *Web-Based Volunteer.* Mentoring middle and high school students to help them toward the engineering field.
**Make-A-Wish Foundation** – *Volunteer.* Fulfill wishes for children diagnosed with critical illnesses. Rebuilt child's backyard to playground in a single day while they went on a special trip with parents.
**Alma Mater Ambassador, Columbia University** – *Ambassador and Interviewer*. Travel to underrepresented areas where Columbia University does not have immediate access to represent the institution and speak about college experience. Interview students applying to the university.
**Columbia University Orchestra** – *Principal Trombonist*. Renowned group that performs in areas such as Lincoln Center and Symphony Space.

## OTHER INFORMATION

**Dual Citizen:** México and United States of America
**Interests:** Eagle Scout with Gold and Bronze Palms, Taekwondo Black Belt Kukkiwon certified in Korea, Varsity Golf Co-Captain, PADI-Certified scuba diver, non-commercial homebrewer.
**Multi-instrumentalist:** Piano, Trombone, Trumpet, Baritone, Tuba, Percussion, Bugle, and Voice.